Physics Final Q-Bank

2 Marks :

QNO:

1) Expand and define the term Laser.

2) Write any four properties of laser.

3) Define induced absorption.

4) Define optical fiber.

5) Define nanoparticles.

6) Expand SEM, TEM, AFM and FTIR.

7) Define heat.

8) Define Crystal.

9) Define stress and strain

10) Define force and momentum.

5 Marks.

Q NO:-

1) Explain the construction and working of He-Ne laser with neat diagram.

2) Explain the types of optical fibers.

3) Explain the properties of carbon nanotubes.

4) Explain the construction and working of SEM with neat diagram.

5) Explain the different types of thermodynamic systems.

6) Explain the types of electromagnetic materials.

7) Differentiate between crystalline and amorphous solids with examples.

8) Mention application of laser and optical fiber.

10 Marks

1) Derive an expression for Einstein's coefficients.

2) Explain the V-I characterstics of zener diode with neat diagram and applications.

3) Derive an expression for interplanar spacing in terms of Miller indices.

4) Calculate the Atomic Packing Factor for ~~BC~~, SC, BCC and FCC.

5) Explain Atomic Force Microscopy (AFM) with neat diagram.

6) Define thermodynamic process and explain different types of thermodynamic process

7) Write a detailed note on thermal and radiation sensors.

8) Write a note on composite materials.

9) Drive an expression for numerical aperture of optical fiber.